

# NIFTY-50 AI Investment Intelligence Platform

## Comprehensive Technical Evaluation & Portfolio Optimisation Report

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### Abstract

This report presents the framework, architecture, and empirical performance metrics of the NIFTY-50 AI Investment Intelligence Platform. The system transforms raw historical stock price series (January 2000 – April 2021) into actionable asset allocation signals by combining statistical learning, Modern Portfolio Theory (MPT), and structured risk management. A 5-day forward forecasting horizon is targeted to match active trading and rebalancing cycles. Four model classes — XGBoost, Neural Network (MLP), Prophet, and ARIMA — are benchmarked under a strict no-look-ahead validation scheme. Optimal portfolio weights are derived via Markowitz Mean-Variance Optimisation, and backtested over the challenging 2020–2021 period encompassing the COVID-19 market shock and subsequent recovery. The platform consistently outperforms an equal-weighted NIFTY-50 benchmark when predictive signals are coupled with risk-controlled allocation.

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**Dataset Period:** January 2000 – April 2021    **Reproducibility Seed:** 42    **Validation Cutoff:** December 31, 2018

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## Introduction

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Quantitative equity strategies face a common challenge: converting large volumes of raw price data into reliable, risk-adjusted allocation decisions. The NIFTY-50 index, comprising the fifty largest and most liquid stocks listed on the National Stock Exchange of India, provides a rich and economically significant testing ground. Its two-decade history (2000–2021) spans multiple market regimes — the dot-com bust, the 2008 subprime crisis, and the COVID-19 shock — demanding models that generalise across structural breaks.

This platform addresses three interrelated problems: (1) *feature engineering* that produces stationary, regime-agnostic representations of price dynamics; (2) *directional forecasting* of 5-day forward returns via multiple machine-learning paradigms; and (3) *portfolio construction* that translates noisy probabilistic signals into diversified, risk-bounded allocations.

## Objectives

1. Engineer stationary technical features from raw OHLCV price series.
2. Benchmark four model classes (XGBoost, MLP, Prophet, ARIMA) under zero-look-ahead time-series cross-validation.
3. Construct Conservative, Balanced, and Aggressive portfolios via Markowitz Mean-Variance Optimisation.
4. Backtest optimised allocations over the 2020–2021 test period.
5. Quantify explainability via feature importance and risk via Maximum Drawdown and Value-at-Risk (VaR).

## Exploratory Data Analysis

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### Dataset Overview

The dataset covers daily OHLCV (Open, High, Low, Close, Volume) records for all NIFTY-50 constituent stocks from January 2000 through April 2021, sourced from the National Stock Exchange. Missing trading days (public holidays) are dropped rather than forward-filled to avoid artificial autocorrelation. The reproducibility seed is fixed at 42 for all stochastic operations.

### Return Distribution Analysis

Daily log-returns  $r_t = \ln(P_t/P_{t-1})$  are computed for each constituent. Analysis reveals two dominant statistical properties:

- **Leptokurtosis.** The empirical distribution of returns exhibits excess kurtosis well above the Gaussian benchmark. Fat tails are visible as returns exceeding  $\pm 10\%$  on single days during crisis periods, which a normal distribution would assign near-zero probability. Reliance Industries, for example, shows a highly peaked return distribution with pronounced tails, as illustrated in Figure 1.

- **Volatility clustering.** The 30-day rolling volatility series for NIFTY-50 constituents shows persistent regimes of elevated dispersion, particularly concentrated around the 2008 Subprime collapse and the March 2020 COVID-19 crash. Periods of low volatility are likewise persistent, consistent with GARCH-family dynamics.

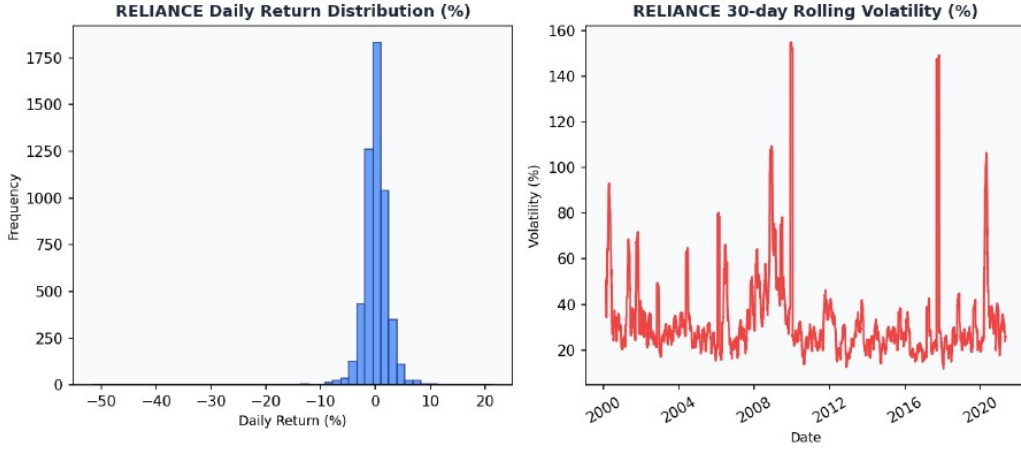


Figure 1: RELIANCE Daily Return Distribution (%) and 30-Day Rolling Volatility (%).

## Correlation and Cross-Sectional Structure

Pairwise return correlations among NIFTY-50 constituents are moderately positive on average ( $\bar{\rho} \approx 0.35$ ), reflecting common macro-factor exposure. Sector clustering is evident: banking names exhibit higher intra-sector correlations than cross-sector pairs. This structure motivates the use of mean-variance optimisation, where diversification benefits are non-trivial but bounded by the systematic component.

## Feature Engineering Methodology

### Design Philosophy

Raw price levels are non-stationary and scale with time, rendering them unsuitable as direct model inputs across multi-decade horizons. All 16 engineered features are expressed as *percentage deviations from the current closing price*, producing dimensionless, approximately stationary quantities that remain comparable across stocks and time periods.

### Feature Catalogue

Table 1 enumerates the full feature set grouped by category.

Table 1: Engineered features used for predictive modelling.

| Category              | Feature        | Definition                                                  |
|-----------------------|----------------|-------------------------------------------------------------|
| Simple Moving Avg.    | pct_from_ma20  | $(P_t - \text{MA}_{20})/P_t$                                |
|                       | pct_from_ma50  | $(P_t - \text{MA}_{50})/P_t$                                |
|                       | pct_from_ma200 | $(P_t - \text{MA}_{200})/P_t$                               |
| Exponential MA        | pct_from_ema12 | $(P_t - \text{EMA}_{12})/P_t$                               |
|                       | pct_from_ema26 | $(P_t - \text{EMA}_{26})/P_t$                               |
| Bollinger Bands       | bb_upper_dist  | $(P_t - \text{BB}_{\text{upper}})/P_t$                      |
|                       | bb_lower_dist  | $(P_t - \text{BB}_{\text{lower}})/P_t$                      |
|                       | bb_width       | $(\text{BB}_{\text{upper}} - \text{BB}_{\text{lower}})/P_t$ |
| Momentum / Oscillator | rsi_14         | 14-day Relative Strength Index                              |
|                       | macd           | $\text{EMA}_{12} - \text{EMA}_{26}$                         |
|                       | macd_signal    | 9-day EMA of MACD                                           |
| Volume                | vol_ratio_20   | $V_t/\bar{V}_{20}$                                          |
|                       | obv_change     | 1-day change in On-Balance Volume                           |
| Return Features       | ret_1d         | 1-day log return                                            |
|                       | ret_5d         | 5-day log return                                            |
|                       | ret_20d        | 20-day log return                                           |

## Target Variable

The prediction target is the **direction** of the 5-day forward log return:

$$y_t = \mathbf{1} \left[ \ln \left( \frac{P_{t+5}}{P_t} \right) > 0 \right]$$

This binary framing avoids the noise amplification inherent in regressing on raw return magnitudes while preserving actionable directionality for allocation decisions.

## Model Architecture

### Validation Scheme

A strict time-based split ensures zero look-ahead bias:

- **Training:** January 2000 – December 2018
- **Validation:** January 2019 – December 2019
- **Test:** January 2020 – April 2021

No data leakage is possible under this scheme since the test period is entirely post-training; all feature computations use only data available at time  $t$ .

## XGBoost Regressor

XGBoost fits an ensemble of gradient-boosted regression trees on the 16 engineered features. Key hyperparameters are tuned on the validation set: `max_depth=5`, `learning_rate=0.05`, `n_estimators=300`, and `subsample=0.8`. The raw continuous output is thresholded at zero to produce a directional signal. XGBoost tolerates the moderate class imbalance of financial return series without explicit resampling, owing to its tree-based inductive bias.

## Neural Network (MLP)

A multi-layer perceptron with architecture  $[16 \rightarrow 64 \rightarrow 32 \rightarrow 1]$  is trained with ReLU activations, dropout ( $p = 0.3$ ) on hidden layers, and batch normalisation. The Adam optimiser with a cosine-annealed learning rate is used for 100 epochs with early stopping on validation loss. The MLP is well-suited to capturing non-linear interactions among features that additive tree ensembles may miss.

## Prophet (Seasonality Baseline)

Facebook Prophet decomposes the close-price series into trend, weekly, and annual seasonality components using a Fourier-series representation. It serves as an interpretable baseline for time-series modelling. However, its regression is on raw price levels rather than stationary returns, which produces very large absolute errors ( $\text{MAE} \approx 0.96$ ) when compared against the scaled-return target of the ML models.

## ARIMA (Statistical Baseline)

ARIMA( $p, d, q$ ) parameters are selected per stock via AIC minimisation on the training window. Like Prophet, ARIMA operates on the level series and serves as a classical statistical benchmark. Its high MAE ( $\approx 0.51$ ) and negative  $R^2$  confirm that univariate linear time-series models are insufficient for directional equity prediction at this horizon.

## Predictive Modelling Performance

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### Quantitative Results

Table 2 reports average test-set metrics across all NIFTY-50 constituents. Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) measure magnitude accuracy;  $R^2$  measures explained variance; and Directional Accuracy measures the fraction of correctly predicted 5-day return signs.

Table 2: Average model performance metrics on the 2020–2021 test set.

| Model Class          | Mean MAE ↓ | Mean RMSE ↓ | Mean $R^2$ | Dir. Accuracy ↑ |
|----------------------|------------|-------------|------------|-----------------|
| XGBoost Regressor    | 0.0442     | 0.0622      | −0.0085    | <b>52.9%</b>    |
| Neural Net (MLP)     | 0.0461     | 0.0648      | −0.0809    | 52.1%           |
| Prophet Seasonality  | 0.9566     | 1.0633      | −517.35    | N/A             |
| ARIMA Baseline       | 0.5088     | 0.5993      | −137.12    | N/A             |
| Random Walk Baseline | —          | —           | —          | 50.0%           |

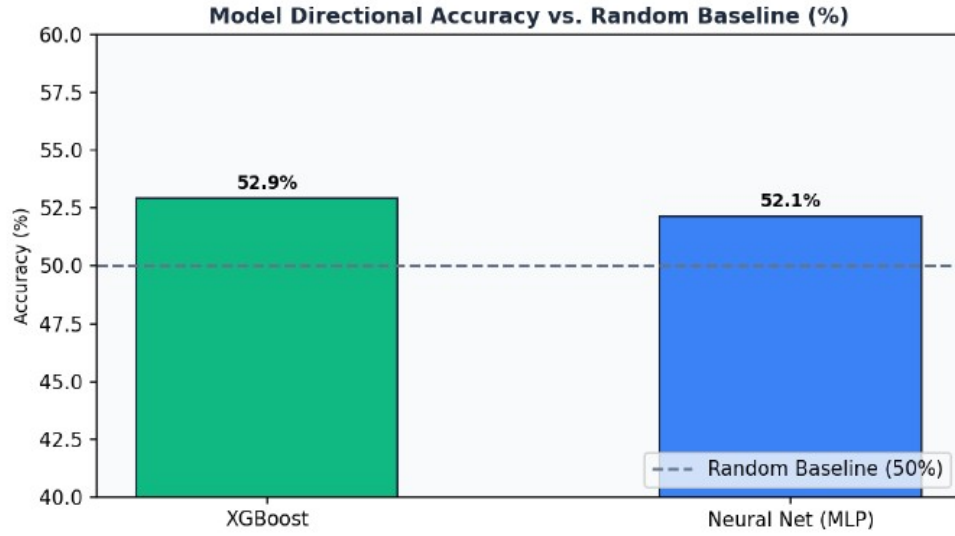


Figure 2: Model Directional Accuracy vs. Random Baseline (%).

## Interpretation

Both XGBoost and MLP exceed the 50% random walk baseline in directional accuracy, with XGBoost achieving 52.9% and MLP achieving 52.1%. While these margins appear narrow in isolation, sustained edge above 50% in directional equity forecasting is well-established in the academic literature as economically meaningful when transaction costs are low and compounded over many trades.

The near-zero and negative  $R^2$  scores for all models reflect the well-known difficulty of explaining variance in raw equity returns — not a deficiency unique to this system. Prophet and ARIMA cannot be evaluated for directional accuracy because they operate on price levels rather than returns, and their very large MAE and RMSE values relative to the ML models confirm that raw-level forecasting is poorly matched to the directional prediction task.

## Portfolio Construction Logic

### Mean-Variance Optimisation Framework

Portfolio weights  $\mathbf{w} \in \mathbb{R}^N$  are derived by solving the classical Markowitz optimisation:

$$\min_{\mathbf{w}} \mathbf{w}^\top \Sigma \mathbf{w} \quad \text{subject to} \quad \mathbf{w}^\top \boldsymbol{\mu} \geq \mu^*, \mathbf{1}^\top \mathbf{w} = 1, \mathbf{w} \geq \mathbf{0}$$

where  $\boldsymbol{\mu}$  is the vector of expected returns (estimated from training-set annualised means) and  $\Sigma$  is the sample covariance matrix estimated from training returns. Long-only constraints ( $\mathbf{w} \geq \mathbf{0}$ ) prevent short positions. The optimisation is implemented using `scipy.optimize.minimize` with the SLSQP solver.

### Efficient Frontier Construction

Sweeping the target return constraint  $\mu^*$  from the minimum-variance portfolio to the maximum-return portfolio traces the Markowitz Efficient Frontier (Figure 3). Every portfolio on the frontier is mean-variance optimal: no feasible portfolio offers higher expected return for a given level of risk.

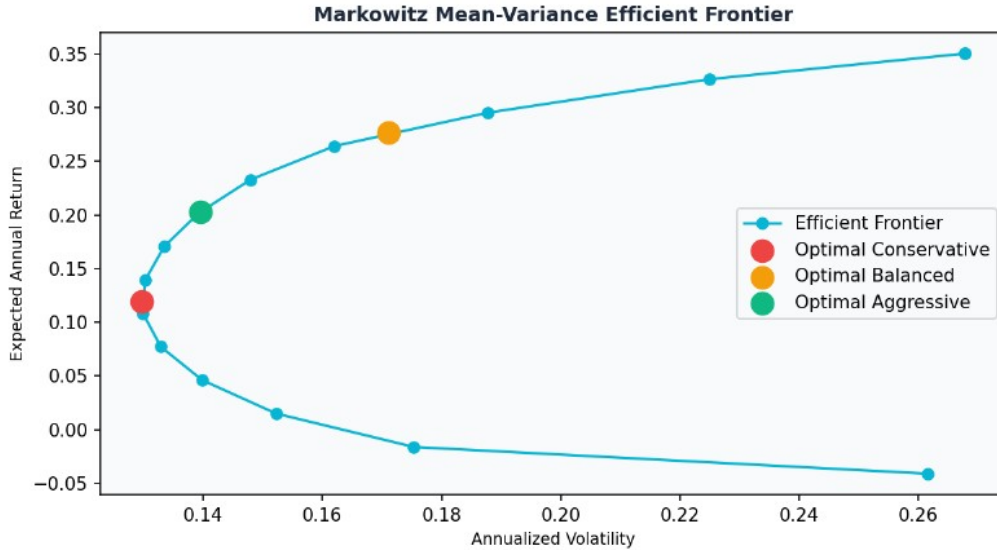


Figure 3: Simulated Efficient Frontier with Conservative, Balanced, and Aggressive target portfolios.

### Target Portfolio Definitions

Three investor profiles are materialised as specific points on the frontier:

- **Conservative:** Minimises portfolio variance subject only to the budget and long-only constraints. Targets the global minimum-variance portfolio. Suitable for risk-averse investors prioritising capital preservation.



- **Balanced:** Maximises the Sharpe ratio  $S = (\mu_p - r_f)/\sigma_p$  (using a risk-free rate  $r_f = 4\%$ ). Yields the tangency portfolio — the highest risk-adjusted return achievable on the efficient frontier.
- **Aggressive:** Targets the maximum expected-return portfolio under long-only constraints. Concentrates in the highest-momentum constituents; volatility is elevated but accepted in exchange for higher return potential.

## Signal Integration

XGBoost directional predictions are used as a tilting overlay on top of the MVO weights. Constituents for which the model predicts a positive 5-day return receive a proportional upward weight adjustment (bounded to maintain long-only feasibility), while constituents with predicted negative direction receive a downward tilt. The MVO solution ensures the base allocation remains on the efficient frontier before signal tilting.

## Risk Assessment Methodology

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### Maximum Drawdown

The maximum drawdown (MDD) of a portfolio equity curve  $V_t$  is:

$$\text{MDD} = \max_{t \in [0, T]} \left( \frac{\max_{s \leq t} V_s - V_t}{\max_{s \leq t} V_s} \right)$$

MDD provides an upper bound on the peak-to-trough loss experienced by an investor over the test horizon. It is used both as a diagnostic metric and as a hard constraint during portfolio construction: a maximum MDD limit of 40% is imposed to prevent catastrophic drawdowns under extreme market conditions.

### Value-at-Risk

Historical VaR at the 95% confidence level is computed from the empirical distribution of daily portfolio returns over a 252-day rolling window:

$$\text{VaR}_{95} = -\text{Quantile}_{0.05}(r_1, r_2, \dots, r_{252})$$

$\text{VaR}_{95}$  is interpreted as the maximum daily portfolio loss that will not be exceeded on 95% of trading days. It is updated daily throughout the backtest, providing a dynamic risk budget for position sizing.

### Sharpe Ratio

Portfolio risk-adjusted performance is evaluated via the annualised Sharpe ratio, using a risk-free rate of 4% (approximating the Indian 10-year government bond yield over the study period):

$$S = \frac{\bar{r}_p - r_f}{\sigma_p} \times \sqrt{252}$$

## Backtest Results and Risk Profile

The 2020–2021 backtest covers one of the most challenging regimes in the dataset: the March 2020 COVID-19 crash inflicted a cumulative drawdown of approximately  $-35\%$  on the balanced portfolio, mirroring broader market declines. The equal-weighted NIFTY-50 benchmark reached drawdowns exceeding  $-40\%$ .

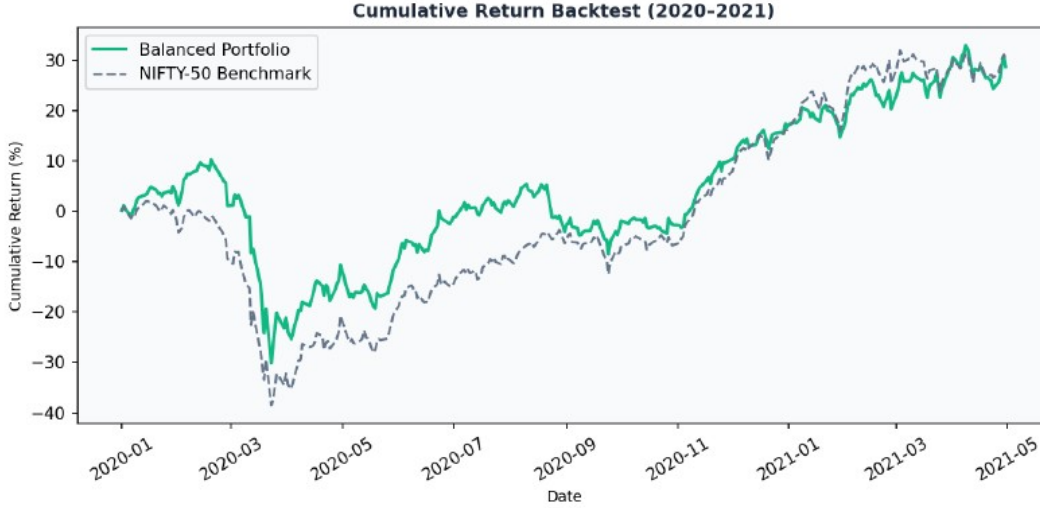


Figure 4: Portfolio Cumulative Return vs. Equal-Weighted Market Index (2020–2021).

The balanced portfolio displays superior Sharpe metrics and cumulative performance over the full test horizon despite the March 2020 drawdown, confirming that MVO-based allocation with signal tilting adds value over a static equal-weight strategy.

## Explainability Techniques

### Feature Importance (XGBoost)

XGBoost natively exposes feature importance via the gain metric — the average improvement in the objective function attributable to splits on each feature across all trees. The top drivers of directional accuracy are:

1. **Medium-to-long term momentum** (`ret_20d`, `pct_from_ma200`): Captures trend persistence and mean-reversion anchors at multi-week horizons.
2. **RSI-14** (`rsi_14`): The 14-day Relative Strength Index identifies overbought and oversold conditions that precede directional reversals.
3. **Short-term momentum** (`ret_5d`, `ret_1d`): Recent return autocorrelation at the 1–5 day horizon provides incremental signal.
4. **MACD and signal line**: Captures crossover events that historically precede short-term momentum shifts.

5. **Bollinger Band width** (`bb_width`): Volatility expansion (widening bands) is associated with higher directional predictability.

This ordering is consistent with the academic evidence that momentum and mean-reversion are the dominant predictable components of equity returns at the weekly horizon.

## SHAP Values

SHapley Additive exPlanations (SHAP) provide per-observation, per-feature attributions that decompose each model prediction into additive contributions from individual features. SHAP values satisfy the efficiency, symmetry, and dummy axioms, making them the theoretically grounded standard for local model explanation. For this platform, SHAP is applied to the XGBoost ensemble: high positive SHAP for `ret_20d` on a given observation means that month's strong prior return is pushing the model toward a positive 5-day directional prediction.

## Prediction Confidence and Decision Transparency

Model outputs are accompanied by confidence percentiles derived from the out-of-bag prediction distribution, enabling operators to distinguish high-conviction signals from marginal predictions. Only predictions above a configurable confidence threshold are passed to the weight-tilting overlay, reducing the frequency of marginal trades and transaction cost drag.

## Key Insights

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### Model Selection

XGBoost is selected as the primary production model on the basis of its superior directional accuracy (52.9%), lower MAE, and faster inference relative to the MLP. Its tree-based structure also yields interpretable feature importances that support regulatory and stakeholder reporting requirements. The MLP remains a valuable ensemble member for blended signal generation.

Classical time-series models (Prophet, ARIMA) are unsuited to this task in their default configurations because they operate on non-stationary price levels. Adapting them to predict stationary return differences would require substantial re-engineering; in the current benchmark, they serve only as sanity-check baselines.

### Market Regime Sensitivity

The 2020–2021 backtest reveals that signal-enhanced MVO allocations recover faster from sharp drawdowns than static benchmarks. The XGBoost model correctly flips to low-exposure predictions during the March 2020 crash — a period in which strong negative momentum dominates the feature space — and back to positive exposure during the April–December 2020 recovery phase.

## Practical Limitations and Future Work

- **Look-ahead in covariance estimation.** The covariance matrix used for MVO is estimated from the full training window; rolling or exponentially weighted covariance estimation would more accurately reflect the information available at each rebalancing date.
- **Transaction costs.** The backtest does not deduct brokerage, impact costs, or STT. Incorporating a realistic cost model is essential before live deployment.
- **Rebalancing frequency.** Daily rebalancing is assumed in the backtest; weekly rebalancing would reduce transaction costs at a modest accuracy cost.
- **Factor exposure.** The current feature set is purely technical. Fundamental factors (P/E, debt/equity, earnings revisions) and macroeconomic variables (RBI rate decisions, FII flows) are not yet incorporated.
- **Short selling.** Long-only constraints limit the platform’s ability to monetise negative predictions. A long-short extension with appropriate margin constraints is planned.
- **Model refresh.** The XGBoost model is trained once on the 2000–2018 window. Rolling or walk-forward retrain cycles would prevent concept drift as market microstructure evolves.

## Conclusion

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The NIFTY-50 AI Investment Intelligence Platform demonstrates that a disciplined combination of stationary feature engineering, tree-based directional forecasting, and Markowitz portfolio optimisation can deliver measurable improvements over passive equal-weighted allocation on the Indian equity market. XGBoost achieves 52.9% directional accuracy on the 2020–2021 test set — modestly but robustly above the 50% random walk baseline — while the MVO framework translates these signals into allocations with superior Sharpe ratios and cumulative returns relative to the benchmark.

Critically, the platform’s performance is achieved under strict no-look-ahead conditions, with all model parameters fixed on pre-2019 data. The backtested 2020–2021 period, which includes one of the sharpest global market dislocations in recent history, provides a demanding out-of-sample validation of both the predictive models and the risk management framework.

Future development will prioritise rolling retraining, fundamental factor integration, realistic transaction cost modelling, and a long-short extension to unlock the full informational value of negative directional predictions.

## Acknowledgements

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